

# CareSync

AI-Powered Remote Healthcare Platform

3-Month Development Plan

| Development Team  |                                       |
|-------------------|---------------------------------------|
| Backend & DevOps: | Ramsey                                |
| Frontend & UI/UX: | Abbie                                 |
| Timeline:         | 3 Months (12 Weeks)                   |
| Target:           | Pilot Launch with Healthcare Agencies |

## Project Overview

CareSync is an AI-powered remote healthcare platform focused on elderly and chronically ill patients. The platform enables continuous care from home through health data logging, remote monitoring, video consultations, and AI-generated insights.

| Core Features       |                                                       |
|---------------------|-------------------------------------------------------|
| Health Data Logging | BP, glucose, weight, temperature, O2 levels, symptoms |
| Remote Monitoring   | Real-time dashboards for doctors and caregivers       |
| Video Consultations | Built-in video calls via Daily.co integration         |
| AI Summaries        | Automated consultation summaries and trend analysis   |

|                    |                                                |
|--------------------|------------------------------------------------|
| Smart Alerts       | Rule-based notifications for critical readings |
| Appointment System | Scheduling and automated email reminders       |

## Technology Stack

| Component | Technology              | Why                                 |
|-----------|-------------------------|-------------------------------------|
| Backend   | Python FastAPI          | Fast, modern, type-safe             |
| Database  | PostgreSQL              | Structured health data, HIPAA-ready |
| Frontend  | Next.js + React         | Modern, fast, SEO-friendly          |
| Styling   | Tailwind CSS            | Rapid UI development                |
| Video     | Daily.co API            | 10k free minutes/month              |
| AI        | OpenAI GPT-4o-mini      | Cost-effective summaries            |
| Hosting   | Railway/Render + Vercel | Free tiers available                |
| Email     | Resend/SendGrid         | Free tier for alerts                |

# MONTH 1: Foundation & Core Features

Weeks 1-4: Authentication, Patient Management, Health Data Logging, Dashboard & Alerts

## Week 1: Setup & Authentication

### Backend Tasks (You):

- Project setup (FastAPI + PostgreSQL)
- Database schema design
- User authentication system (JWT)
- Multi-tenant setup (agency isolation)
- User roles: Admin, Doctor, Caregiver, Patient
- Basic CRUD endpoints for users
- Docker setup for local development
- Environment configuration

### Frontend Tasks (Abigael):

- Next.js project setup
- Tailwind CSS configuration
- Design system/component library (shadcn/ui)
- Login/Register pages
- Basic routing structure
- API client setup (axios/fetch)
- Authentication state management

**Deliverable:** Working auth system with role-based access

## Week 2: Patient Management

### Backend Tasks (You):

- Patient model & database schema
- CRUD endpoints: Create, List, Get, Update, Delete patients
- Patient-caregiver-doctor relationships
- Pagination & filtering
- Input validation (Pydantic models)

### Frontend Tasks (Abigael):

- Patient list page (dashboard)
- Add new patient form

- Patient detail view
- Edit patient form
- Search & filter UI
- Responsive mobile design (caregivers use phones!)

**Deliverable:** Complete patient management (CRUD)

## Week 3: Health Data Logging

### Backend Tasks (You):

- Health readings model (BP, glucose, weight, temp, O2, heart rate)
- Flexible schema for different reading types
- Endpoints: Log readings, Get history, Get latest vitals
- Photo/file upload (medication bottles, wounds)
- Validation rules (BP range, glucose range)
- Trend calculation logic

### Frontend Tasks (Abigael):

- Health data entry form (quick & detailed modes)
- Photo capture/upload
- Reading history view
- Trend graphs using Recharts (BP, glucose, weight over time)
- Mobile-optimized input

**Deliverable:** Health logging with graphs

## Week 4: Dashboard & Alerts

### Backend Tasks (You):

- Alert system architecture
- Rule-based alerts (high/low BP, glucose, missed readings)
- Alert endpoints (List, Acknowledge)
- Email notification system (SendGrid/Resend)
- Alert prioritization (critical, warning, info)

### Frontend Tasks (Abigael):

- Doctor/Caregiver dashboard with patient list
- Alert feed (critical alerts highlighted)
- Quick stats (total patients, alerts today, consultations)
- Alert notification UI (toast notifications, alert center)
- Patient status cards (healthy, needs attention, critical)
- Dashboard filters (by risk level, last contact)

**Deliverable:** Complete dashboard with real-time alerts

## MONTH 2: Video & AI Features

Weeks 5-8: Video Consultations, AI Summaries, Scheduling, Admin Panel

## Week 5: Video Consultations

### Backend Tasks (You):

- Daily.co API integration
- Consultation scheduling model
- Endpoints: Schedule, List, Start (generate room), Complete
- Store consultation metadata (duration, participants, notes)
- Recording webhook handler (if using Daily.co recording)

### Frontend Tasks (Abigael):

- Schedule consultation form (date/time picker, select patient/doctor)
- Consultation list view (upcoming & past)
- Video call UI (embed Daily.co, in-call vitals display)
- End call button
- Mobile video optimization

**Deliverable:** Complete video consultation flow

## Week 6: AI Summaries (YOUR DIFFERENTIATOR!)

### Backend Tasks (You):

- OpenAI API integration (GPT-4o-mini)
- Consultation summary generation (notes + vitals → summary)
- Weekly health summary (week of data → trend analysis)
- Endpoints: Generate consultation summary, Weekly report
- Prompt engineering for medical context
- Error handling & retry logic
- Summary storage in database

### Frontend Tasks (Abigael):

- Consultation summary view (display AI summary after call)
- Edit/annotate summary, Save final version
- Weekly report view (auto-generated, trend highlights)
- AI loading states
- Summary history

**Deliverable:** AI summary generation working

## Week 7: Scheduling & Reminders

### Backend Tasks (You):

- Appointment scheduling system
- Doctor availability management
- Endpoints: Create, List, Reschedule, Cancel appointments
- Automated reminder system (cron job: 24hrs & 1hr before)
- Email reminders (no SMS yet - too expensive)
- Daily health check-in reminders

### Frontend Tasks (Abigael):

- Appointment booking flow (calendar view, time slot selection)
- Appointment management (reschedule, cancel)
- Reminder preferences settings
- Calendar integration (export to Google Calendar)

**Deliverable:** Scheduling with automated reminders

## Week 8: Admin Panel & Agency Management

### Backend Tasks (You):

- Agency (tenant) management
- Endpoints: Create agency, Get stats, List patients/users
- Billing/usage tracking (patients, consultations, AI summaries)
- Analytics endpoints (patient growth, consultation frequency, alerts)

### Frontend Tasks (Abigael):

- Admin dashboard (agency overview, user management)
- Analytics views (charts for growth, consultations, alerts)
- Usage metrics (for billing discussions)
- Settings panel

**Deliverable:** Admin panel for agencies

## MONTH 3: Polish, Testing & Pilot

Weeks 9-12: Optimization, Testing, Deployment, Launch

## Week 9: Mobile Optimization & UX Polish

### Backend Tasks (You):

- API performance optimization (indexing, query optimization, caching)
- Rate limiting
- API documentation (FastAPI auto-docs + examples)
- Error handling improvements
- Logging & monitoring setup

### Frontend Tasks (Abigael):

- Mobile responsiveness audit (test on real devices)
- Progressive Web App (PWA) setup (offline, add to home screen)
- UX improvements (loading states, error messages, tooltips)
- Accessibility audit (basic)

**Deliverable:** Mobile-optimized, polished UI



## Week 10: Testing & Bug Fixes

### Backend Tasks (You):

- Unit tests (auth, alerts, AI summaries)
- Integration tests (API endpoints, database)
- Security audit (SQL injection, XSS, CORS, encryption)
- Load testing (basic)
- Bug fixes from frontend testing

### Frontend Tasks (Abigael):

- Manual testing across all flows
- Cross-browser testing
- Mobile device testing
- Create bug list & fix critical bugs
- User acceptance testing (UAT with friends/family)

**Together:** End-to-end testing, test scenarios, integration bug fixes

**Deliverable:** Tested, bug-free system

## Week 11: Deployment & Documentation

### Backend/DevOps Tasks (You):

- Production database setup (PostgreSQL on Railway/Render/Supabase)
- Environment variables management
- Backend deployment (Railway/Render)
- SSL/HTTPS setup
- Database backups setup
- Monitoring (Sentry free tier)
- CI/CD pipeline (GitHub Actions - basic)
- API documentation finalization

### Frontend Tasks (Abigael):

- Frontend deployment (Vercel)
- Environment configuration (API URLs)
- Custom domain setup (if available)
- Performance optimization (images, code splitting, lazy loading)
- Analytics setup (PostHog/Plausible free tier)
- User documentation/help section

**Together:** Full system smoke test, demo account with sample data, pitch deck/demo video

**Deliverable:** Deployed system on custom domain

## Week 12: Pilot Launch & Iteration

### Your Tasks:

- Onboard first pilot agency/hospital
- Setup their account & users
- Train administrators
- Monitor system performance
- Quick bug fixes
- Collect feedback
- Plan iteration 2

### Abigael's Tasks:

- Create onboarding materials (videos, guides, FAQ)
- User support (respond to questions)
- Track user behavior (most-used features)
- UI tweaks based on feedback
- Iterate on confusing flows

**Together:** Weekly pilot review meetings, document learnings, prioritize iteration 2

**Deliverable:** Live pilot with real users!

## Success Metrics

| Month   | Goals                                                                                                                                                      |
|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Month 1 | <ul style="list-style-type: none"><li>• Complete auth &amp; patient management</li><li>• Health data logging works end-to-end</li></ul>                    |
| Month 2 | <ul style="list-style-type: none"><li>• Video calls working</li><li>• AI summaries generating</li><li>• Alert system functional</li></ul>                  |
| Month 3 | <ul style="list-style-type: none"><li>• Live pilot with 1 hospital/agency</li><li>• 20-50 patients in system</li><li>• Feedback collected for v2</li></ul> |

## Tools & Workflow

- **Project Management:** GitHub Projects or Trello (free) - Task tracking
- **Version Control:** GitHub - Code repository
- **Communication:** Discord/Slack - Daily communication
- **Design:** Figma (free) - UI mockups
- **Documentation:** Notion/Google Docs
- **API Testing:** Postman/Insomnia
- **Development:** VS Code, Docker for local environment

## Daily Routine

**Morning:** 15-min async standup (text update on Discord/Slack)

- What you did yesterday
- What you're doing today
- Any blockers

**Evening:** Push code, update task board

**Weekly:** 1-hour sync call (discuss progress, plan next week)

## Budget Breakdown (Monthly)

| Service | Free Tier | Cost When Scaling |
|---------|-----------|-------------------|
|---------|-----------|-------------------|

|                                          |                                        |                                                  |
|------------------------------------------|----------------------------------------|--------------------------------------------------|
| Daily.co (video)                         | 10k min/month                          | \$24/month (100 patients)                        |
| OpenAI (AI)                              | Pay-as-you-go                          | ~\$3-6/month (MVP)                               |
| Railway/Render (backend)                 | Free tier                              | \$5-10/month                                     |
| Vercel (frontend)                        | Free tier                              | Free (hobby)                                     |
| Supabase (database)                      | 500MB free                             | \$25/month (growth)                              |
| SendGrid (email)                         | 100/day free                           | Free for MVP                                     |
| <b>&lt;b&gt;Total MVP Cost&lt;/b&gt;</b> | <b>&lt;b&gt;\$0-10/month&lt;/b&gt;</b> | <b>&lt;b&gt;\$60-75/month (growth)&lt;/b&gt;</b> |

## Risk Mitigation Strategies

| If Behind Schedule... | Mitigation Strategy                                                                      |
|-----------------------|------------------------------------------------------------------------------------------|
| Weeks 1-4 behind      | Skip admin panel (Week 8), focus on core patient features                                |
| Weeks 5-7 behind      | Simplify AI summaries (use templates instead of full AI)                                 |
| Weeks 9-12 behind     | Launch with known bugs, fix iteratively with pilot feedback                              |
| Frontend struggles    | Pair programming on complex components, use UI templates (shadcn/ui)                     |
| Money runs out        | Daily.co free tier → WebRTC<br>Reduce AI summary frequency<br>Use all free hosting tiers |

## Pro Tips for Success

- **Start with the hardest part first** - Video + AI in Month 2 prevents last-minute crunch
- **Deploy early** - Don't wait till Week 11, deploy by Week 4 to catch issues early
- **Get real users by Week 8** - Even 5 patients from family/friends for testing
- **Document as you go** - Future you will thank present you
- **Don't over-engineer** - Ship features quickly, iterate based on feedback
- **Communicate daily** - 15-min async updates prevent blockers
- **Celebrate small wins** - Deployed auth system? Celebrate! First video call? Celebrate!
- **Ask for help** - Stuck on something for >2 hours? Reach out to communities (Discord, Reddit)

## Next Steps

1. Review this plan together and discuss timeline feasibility
2. Set up communication channels (Discord/Slack)
3. Create GitHub repository and project board
4. Design database schema (Week 1, Day 1)
5. Start coding! ■

*Development Plan Generated: May 11, 2026*